

A Packing of 16 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 16 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.844074. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.832994, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 16 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.844074 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 16$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.832994, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 16/15 = 2.844444$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N16.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
142203731913939832957013190389814150701454647341542119453038056535880076
907810954043771530980540269411916158319393520035074583575824581683837638
6680888204844763776853963297261464815940379806033
```

denominator:

```
4999999999999442322687620468342118176564718213400472943274234137469759704
118124434926456693522351523542164607428310784953642684111057549053589997
762598343397284608836053422028713618007544001493
```

B Coordinates

16 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N16.txt`.

```
-61665867732 -231384448030 -176623975460 1192615228950255380634 -466448097193 -466376808058 9070676336662632902048 -133557095594 -358802022814 -487268665006
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